

FERMENTFREUDE

Booking, Rebooking & Refund Logic

Business Rules and Architecture Reference

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Version 1.1 | 14 July 2026

Payload CMS • Stripe • Brevo • Workshop Dashboard

Purpose: Translate Fermentfreude's approved booking and cancellation policy into clear business outcomes, decision rules and acceptance criteria. Suggested status names, fields and pseudocode are reference patterns for developer assessment, not mandatory implementation instructions.

Document Control

Field	Value
Document owner	Fermentfreude OG - David Haider and Marcel Raunnigger
Primary users	Rafaela Vaz, Ala'a Shaheen, future Fermentfreude developers
Version	1.1
Effective architecture baseline	14 July 2026
Related systems	Payload CMS, Stripe, Brevo, existing product + appointment model
Legal source of truth	The AGB version accepted for the individual booking
Business source of truth	Approved business outcomes in this specification and later explicit written decisions by Fermentfreude. The technical model and naming remain subject to developer assessment.
Review status	Business policy approved; technical reference model for developer assessment; final legal text remains subject to external legal review.

Critical implementation principle: Thresholds, capacities, rebooking groups and legal text versions must be configurable. Do not hard-code €99, workshop names, or current capacities into cancellation and rebooking logic.

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1. Purpose and Scope

The Fermentfreude website is a live operational platform, not only a marketing website. This specification defines how the booking lifecycle must behave across the dashboard, customer self-service, workshop capacity, Stripe payments and refunds, Brevo transactional emails, vouchers and internal history.

- **In scope:** fixed-date workshop bookings, cancellations, rebookings, refunds, replacement participants, multi-seat bookings, organiser cancellations, capacity exceptions, vouchers used as payment and relevant technical failure cases.
- **Out of scope for the first implementation:** automatic account-based credit balances, waitlists, course-series cancellation logic, fully automated different-price self-rebooking and shipping operations.
- **Architecture goal:** implement a stable booking lifecycle now without preventing future workshop types, prices, event formats or fulfilment methods.

Where the current codebase supports the required outcomes through a simpler or safer design, Rafaela and Ala'a should propose that alternative and document any deviation. Fermentfreude should approve changes to business outcomes; developers retain responsibility for the implementation approach.

Business outcomes and customer rights are requirements. Status names, field names, collection structure, event names and pseudocode are illustrative unless explicitly described as a business rule.

How to Use This Document

2. Guiding Principles and Source-of-Truth Hierarchy

1. The AGB version accepted for a booking determines the customer's contractual rights for that booking.
2. This document defines the required business outcomes and provides a technical reference model for implementing the approved policy.
3. New AGB or policy versions must never be applied retroactively to older bookings.
4. Customer self-service must be rule-based and limited. Admin actions must remain flexible but fully auditable.
5. The implementation must keep booking/entitlement, capacity, payment, refund, voucher, attendance and communication states distinguishable, whether through separate fields, entities or an equivalent design.
6. No financial action may be inferred from silence. In particular, an organiser cancellation must not automatically convert payment into a voucher or a specific new date.
7. Every webhook and financial operation must be idempotent and safely repeatable.

Legal/technical boundary: This specification does not replace the AGB or withdrawal information. The system must store and apply the correct terms version and display the legally relevant checkout information for the relevant product category.

3. Domain Terminology and Conceptual Model

Term	Meaning in this specification
Workshop product / workshop type	Reusable format and content, e.g. Tempeh, Kombucha or Lacto-Fermentation.
Appointment / workshop date	One concrete bookable date with start/end time, location, capacity and status.
Order	The commercial checkout transaction. One order can contain multiple seats and payment components.
Booking	The relationship between an order/customer and one appointment.
Booking unit / seat	One individual paid participation entitlement. Multi-seat bookings contain multiple seat records or an equivalent seat-level representation.
Regular workshop	A customer-facing workshop eligible for normal self-service rebooking. Defined by configuration, not by name or price literal.
Special / partner event	An event that may have different pricing, location or conditions and may be excluded from customer self-service.
Self-rebooking	A customer-initiated one-time change under the standard rules.
Admin rebooking	A manual change by David or Marcel. May target any event or appointment and may override normal restrictions.
Rebooking entitlement	A right linked to an existing paid seat. It is not a generic voucher or account credit.
Purchased value voucher	A voucher bought for money, with a stored monetary balance.
Promotion / goodwill voucher	A free voucher or discount. It is not cash-redeemable.
Refund	Return of the eligible actually-paid amount to the original payment source, subject to voucher restoration rules.

Recommended model: Use one Order with one or more seat-level Booking Units. Seat-level records are strongly preferred because partial cancellation, rebooking, replacement and no-show status must be possible without changing the remaining seats.

4. Approved Business Rules

4.1 Booking acceptance and terms

BR-001 Booking acceptance

A successful Stripe payment alone is not sufficient to create a confirmed participation entitlement. The platform must record Fermentfreude's acceptance and make the booking confirmation available to the customer, for example on-screen and/or by email. A later email-delivery failure does not by itself reverse an otherwise validly confirmed booking.

BR-002 Payment without booking

If payment succeeds but the booking cannot be accepted, the system must create an exception requiring review. If no seat can be provided, the customer may choose rebooking or full refund.

BR-003 Terms version

Every order must store the AGB version, cancellation-policy version, acceptance timestamp and a stable reference or snapshot of the accepted text.

4.2 Customer cancellation matrix

Time before original workshop start	Customer entitlement	System result
At least 720 hours (30 days)	Full refund OR one self-rebooking	REFUND_REQUESTED or SELF_REBOOK_FLOW
At least 336 hours, less than 720 hours	One self-rebooking; no contractual cash refund	SELF_REBOOK_FLOW
Less than 336 hours (14 days)	No refund, credit or regular rebooking	CANCELLED_NO_REFUND
No-show	No refund, credit or rebooking	NO_SHOW

BR-010 Time calculation

The decision is calculated from the exact published appointment start timestamp in the Europe/Vienna time zone. 30 days = 720 hours and 14 days = 336 hours.

BR-011 Late cancellation notification

A customer may still notify Fermentfreude that they will not attend inside the 14-day window. The seat is marked CANCELLED_NO_REFUND and can be released, but no financial or rebooking entitlement is created.

BR-012 Personal reasons

Illness, work commitments, childcare, travel problems, confusion about the date or a missing reminder do not automatically change the standard outcome.

BR-013 Goodwill

Admins may grant a different outcome as goodwill. Goodwill is not a customer entitlement and must be recorded with actor, timestamp, reason and financial effect.

Once a customer confirms a cancellation and the seat is released, the cancellation is final in self-service. A replacement participant must be entered instead of cancelling if the customer wishes to keep the paid seat. Any reversal after cancellation requires an admin action.

BR-014 Final cancellation and seat release

4.3 Customer self-rebooking

BR-020 One-time rule

A seat may be self-rebooked once only. After a completed customer rebooking, no further self-rebooking or customer-initiated cash refund right is created.

BR-021 Eligibility

Self-rebooking is allowed only while the original appointment is at least 336 hours away, except for the short-notice correction window in BR-025.

BR-022 Target eligibility

The target must be available and configured as eligible under the applicable self-rebooking policy. Eligibility may be represented by a rebooking group, price/equivalence rule or another maintainable configuration. The code must not compare against a hard-coded €99 value or workshop name.

BR-023 Equivalent workshops

For the current standard workshops, customers may rebook between Tempeh, Kombucha and Lacto workshops when they are configured as equivalent regular workshops. The implementation should support configurable equivalence and price handling rather than relying on names or a fixed amount.

BR-024 No new rights after rebooking

The original booking date, original eligibility and rebooking history remain stored. Rebooking must not reset cancellation deadlines or generate a new one-time self-rebooking entitlement.

BR-025 Short-notice correction window

If the original booking was made less than 336 hours before start, the customer may correct an obvious wrong-date or wrong-workshop booking within 24 hours of confirmation, provided the workshop is still at least 48 hours away. This counts as the one self-rebooking and does not create a refund right.

4.4 Replacement participants and multi-seat bookings

BR-030 Replacement person

Any paid seat, including an individual booking, may be transferred to another participant free of charge until workshop start. The payment and booking remain linked to the original buyer.

BR-031 Multi-seat independence

Cancellation, rebooking, replacement, refund or no-show must be executable for one or more individual seats without cancelling or changing the remaining seats.

BR-032 Participant details

Seat IDs must exist even if individual participant names were not collected at checkout. Names and allergy information may be added or updated later.

4.5 No-show, lateness and reminders

BR-040 No-show

No-show creates no refund, credit or rebooking entitlement. An admin can still override as goodwill.

BR-041 Late arrival

The dashboard may record late arrival or denied entry. There is no automatic refund or extension. Operational handling remains with the workshop team.

BR-042 Reminders

Reminder emails are a service. Failed or missing reminders do not invalidate a confirmed booking. The system must nevertheless record email delivery/failure status.

4.6 Cancellation or major change by Fermentfreude

BR-050 No automatic new date

When Fermentfreude cancels an appointment, customers must not be automatically moved to another date. The original appointment and affected seats are marked as cancelled by organiser.

BR-051 Customer options

Each affected seat retains the choice between selecting any available regular workshop configured for organiser-cancellation rebooking and requesting a full refund of the currently refundable amount linked to that seat. If a target has a different price, the implementation must support an explicit price-handling step or an admin-assisted fallback.

BR-052 No mandatory credit

The platform must not automatically create an account credit or voucher. Silence is not acceptance of a voucher or a new date.

BR-053 Long-lived entitlement

The rebooking/refund entitlement is not subject to a short contractual expiry. A magic link may expire for security, but the entitlement must remain and a replacement link must be issuable.

BR-054 Refund request timing

After the customer requests refund, Fermentfreude must initiate it without undue delay and no later than 14 days after the valid request.

BR-055 Repeat organiser cancellation

If a rebooked replacement appointment is later cancelled or materially changed by Fermentfreude, the customer receives the organiser-cancellation choices again.

Selecting a replacement after an organiser cancellation is not the customer's voluntary one-time self-rebooking. It must not increment or reset self_rebooking_count. Any previously used self-rebooking count remains unchanged.

BR-056 Organiser-cancellation rebooking does not consume the customer allowance

4.7 Minor and major changes by Fermentfreude

Change type	Default classification	Customer right
Start-time shift up to and including 60 minutes	Minor, unless objectively unreasonable in the specific case	Information; no automatic refund
Start-time shift over 60 minutes	Major	Accept change, rebook or full refund
Different calendar day	Major	Accept change, rebook or full refund
Materially different location, duration or core content	Major	Accept change, rebook or full refund
Qualified facilitator change / minor flow adjustment	Minor	Information; admin goodwill remains possible

4.8 Minimum participants and appointment cancellation

BR-060 Minimum participant signal

The default minimum is four paid seats. This is a dashboard decision signal, not an automatic cancellation trigger.

BR-061 Manual decision

Fermentfreude may run a workshop with fewer than four participants. Only an admin may cancel due to low participation.

BR-062 Low-participation deadline

Under the approved policy, a cancellation solely because the minimum has not been reached must be decided and communicated no later than 48 hours before start.

4.9 Refund execution and calculation

BR-070 Semi-automated refund

The system calculates the policy result and refund amount. Stripe refund execution requires admin confirmation for the MVP.

BR-071 Admin confirmation is operational only

For a contractually due refund, the admin step verifies amount, seat, payment source and technical consistency. It is not a discretionary approval of whether the customer receives the due refund.

BR-072 Actually-paid amount

Refunds are calculated from the current refundable financial components linked to the affected seat, including the original allocation and any later additional payment or partial refund resulting from rebooking. Never calculate from the current list price alone or overwrite the original transaction history.

BR-073 No fee deduction

Do not deduct Stripe or processing fees from the customer's eligible refund.

BR-074 Original payment sources

Card/payment-service amounts are refunded to the original payment source. Purchased voucher value is restored to the voucher. Free discount/goodwill value is not paid out in cash.

4.10 Admin rebooking and override

BR-080 Admin rebooking scope

Admins may move a customer to any appointment or event, including special, partner, private or unlisted events, subject to capacity and an explicit price decision.

BR-081 Price difference handling

For admin rebooking, the dashboard must show original paid amount, target price and difference. Admin may request additional payment, waive it as goodwill, issue a partial refund for a cheaper target, or document another individually agreed outcome.

BR-082 Capacity override

Admins may use additional capacity up to the configured maximum capacity. Above the maximum, the system should block confirmation unless the maximum itself is explicitly changed by an authorised admin.

BR-083 Audit requirement

Every override must record reason, actor, timestamp, old and new values, financial effect and communication status.

4.11 Voucher and product classification**BR-090 Purchased voucher**

A purchased value voucher is a payment instrument with stored monetary balance. When a booking is unwound, the used purchased-voucher value is restored rather than paid as cash.

BR-091 Free voucher

A promotion or goodwill voucher is not cash-redeemable. Its reactivation after cancellation depends on its own rules and validity.

BR-092 Rebooking entitlement is not a voucher

The right created by organiser cancellation must remain linked to the original seat/payment and must not be implemented as a generic discount code unless the implementation preserves all original rights and traceability.

BR-093 Checkout legal classification

The system must distinguish `FIXED_DATE_WORKSHOP`, `PURCHASED_VOUCHER`, `PERISHABLE_GOOD` and `WITHDRAWABLE_GOOD` because each category has different customer information and withdrawal handling.

If an admin moves a customer to an event with materially different event-specific conditions, the customer must receive those conditions and explicitly agree before the change is finalised. The original AGB/version history remains preserved.

BR-084 Different event-specific conditions**5. Customer and Admin Flows****5.1 Customer cancels 30+ days before start**

8. Customer opens the secure booking-management link and selects one or more seats.
9. System calculates ≥ 720 hours and shows two choices: full refund or one-time rebooking.
10. Refund choice creates `REFUND_REQUESTED` with the calculated amount. Seat is removed from active capacity according to the cancellation transaction.
11. Admin confirms refund execution; Stripe/voucher restoration is triggered.
12. Rebooking choice opens eligible appointments and completes the seat transfer atomically.
13. Brevo sends the appropriate confirmation; audit history is written.

5.2 Customer cancels between 14 and 30 days

14. System shows one-time rebooking and replacement-person options.
15. Cash refund is not presented as a standard entitlement.
16. If customer chooses "I will not attend", seat becomes `CANCELLED_NO_REFUND` and may be released.
17. Admin can later apply goodwill without changing the original policy result.

5.3 Customer cancels less than 14 days before start

18. System clearly states: no refund, no standard rebooking and no credit.
19. Customer may report non-attendance or provide a replacement participant.
20. Reported non-attendance is `CANCELLED_NO_REFUND`, not `NO_SHOW`.
21. Any later rebooking/refund/voucher is an explicit admin goodwill action.

5.4 Fermentfreude cancels a complete appointment

22. Admin changes appointment to `CANCELLED_BY_ORGANISER` and enters customer-facing reason plus internal note.
23. All affected active seats move to `ORGANISER_CANCELLED_DECISION_PENDING`. Payment remains linked.
24. Brevo sends each buyer a secure link and explains both options: select an available regular workshop configured for organiser-cancellation rebooking, or request full refund. Different-price or special-event choices may use an admin-assisted path until automated price handling is available.

- 25. No customer is automatically assigned to a new date and no generic credit is created.
- 26. Customer selection updates each seat independently. Multi-seat orders may choose different outcomes per seat.
- 27. If no choice is made, entitlement remains open. Admin may remind or, after notice, refund proactively.
- 28. Acceptance, rebooking or refund is recorded per seat and triggers the corresponding confirmation.
- 29. The customer can accept the change, choose an eligible replacement workshop, or request full refund.
- 30. Affected seats are marked as awaiting a decision without automatically changing the booking.

5.5 Customer response to a major organiser change

5.6 Admin moves a customer to a special event

- 31. Admin selects the affected seat and any target appointment, including event workshops.
- 32. Dashboard displays target availability, regular/max capacity, price difference, original payment and prior changes.
- 33. Admin selects financial handling and documents the reason.
- 34. If additional payment is required, target seat should be held with an expiry until payment succeeds.
- 35. The old seat is released and the target seat confirmed in one auditable operation.
- 36. A new confirmation email is sent with target date, location and price handling. If the target has materially different event-specific conditions, explicit customer acceptance is required and the accepted conditions/version are stored.

6. Reference Status Model and State Transitions

Reference-model rule: Keep lifecycle, visibility, availability, financial and attendance concepts distinguishable. The following names are examples; developers may use an equivalent model that preserves the same business outcomes and auditability.

6.1 Appointment status

Domain / suggested value	Meaning
Lifecycle: DRAFT	Not ready for customers.
Lifecycle: PUBLISHED	Operationally active; bookability still depends on visibility, sales window and capacity.
Lifecycle: CANCELLED_BY_ORGANISER	Will not take place.
Lifecycle: COMPLETED	Workshop took place.
Lifecycle: ARCHIVED	Historical/non-operational record.
Visibility: PUBLIC / UNLISTED / HIDDEN	Separate control for where and how the appointment is shown.
Availability: AVAILABLE / SOLD_OUT / CLOSED	Preferably derived from released capacity, sales window and lifecycle rather than used as the sole lifecycle status.

6.2 Seat / booking-unit status

Suggested entitlement status	Meaning
PENDING_PAYMENT	Temporary entitlement/hold while payment is unresolved.
ACTIVE	Confirmed participation entitlement.
CANCELLED_BY_CUSTOMER	Customer ended the entitlement; refund outcome is tracked separately.
REBOOKING_PENDING	Source/target transition or price handling is incomplete.
REBOOKED	Original entitlement transferred and linked to a target entitlement.
ORGANISER_CANCELLED_DECISION_PENDING	Appointment cancelled; customer choice is outstanding.
CANCELLED_BY_ORGANISER_RESOLVED	Organiser cancellation has been resolved through rebooking or refund.

6.3 Payment and refund status

Domain	Suggested statuses
Payment	REQUIRES_PAYMENT, PROCESSING, PAID, FAILED, PARTIALLY_REFUNDED, REFUNDED, DISPUTED
Refund	NOT_REQUESTED, REQUESTED, APPROVED_FOR_EXECUTION, PROCESSING, PARTIALLY_COMPLETED, COMPLETED, FAILED, CANCELLED
Voucher	ACTIVE, PARTIALLY_REDEEMED, REDEEMED, EXPIRED, DISABLED, RESTORED
Attendance	NOT_MARKED, ATTENDED, NO_SHOW, LATE, ENTRY_DENIED
Participant identity	Current participant details plus replacement history/event; replacement need not be a lifecycle status.
Email	QUEUED, SENT, DELIVERED_IF_AVAILABLE, FAILED, RESENT

6.4 Key transition constraints

- A completed self-rebooking increments self_rebooking_count and permanently disables another self-service rebooking for that seat.
- Rebooking must link original_seat_id and target_seat_id; do not delete the original record.
- Refund completion must not be inferred from a local status alone; reconcile with Stripe refund status.
- Capacity must be derived from capacity-consuming seat states, not order count.
- Replacement participant changes participant identity, not payment or booking ownership; represent it as data/history rather than forcing a new entitlement lifecycle state.
- No-show is an attendance outcome set only after the appointment starts/passes. Customer-notified non-attendance ends the entitlement as a customer cancellation without refund; financial status remains separate.

7. Required Capabilities and Reference Data Model

7.1 Workshop product / type

Suggested field / capability	Purpose
id, title, slug	Identity and website content.
event_type	REGULAR_WORKSHOP, SPECIAL_EVENT, PARTNER_EVENT, PRIVATE_EVENT, ONLINE_EVENT, COURSE_SERIES.
rebooking_group_id	Defines customer self-service equivalence. Field name and representation are implementation choices.
price_category_id	Configurable price/equivalence class or policy input; must not be a hard-coded euro value.
self_rebooking_eligible	Whether appointments of this type may appear in customer self-service.
organiser_cancel_rebooking_eligible	Whether it may be selected after Fermentfreude cancellation.
default_regular_capacity / default_max_capacity	Defaults copied to new appointments.
default_minimum_participants	Default 4, but configurable.

7.2 Appointment / concrete date

Suggested field / capability	Purpose
id, workshop_type_id	Links date to reusable workshop product.

Suggested field / capability	Purpose
start_at, end_at, timezone	Authoritative eligibility and communication timestamps.
location_id / location_text	Customer-facing venue.
status, visibility	Operational and website state.
regular_capacity	Online self-service capacity.
maximum_capacity	Absolute/admin-reviewed limit.
additional_seats_released	How many beyond regular capacity are currently offered.
minimum_participants	Decision signal, normally 4.
cancellation_reason_customer / internal_note	Separate public and internal text.
cancelled_at / cancelled_by	Audit.

7.3 Order and booking unit / seat

Entity	Required fields
Order	id, buyer/customer, created_at, currency, total, discount allocation, voucher allocation, Stripe IDs, terms_version, accepted_at.
Booking / seat	id, order_id, appointment_id, entitlement status, participant details, original pricing allocation, linked financial adjustments/ledger, self_rebooking_count, original/target linkage, cancellation timestamp/reason.
Attendance	Attendance outcome stored separately from entitlement status; recorded_by and recorded_at.

7.4 Financial and operational records

Entity	Required fields
Payment	provider, PaymentIntent/Charge IDs, amount, currency, status, paid_at, idempotency keys.
Refund	seat IDs, policy result, requested amount/components, execution amount/components, reason, provider refund IDs, separate component status, requested/confirmed/completed timestamps, actor.
Rebooking transaction	source seat, target appointment/seat, type SELF/ADMIN/ORGANISER_CANCEL, original eligibility snapshot, price handling, additional payment/refund references, accepted event-specific terms if applicable, actor, timestamps.
Voucher transaction	voucher ID, value before/after, redeemed/restored amount, related order/seat/refund.
Communication log	template/event type, recipient, Brevo message ID, status, timestamps, resend history.
Audit event	actor, action, entity, before/after summary, reason, IP/session where appropriate, timestamp.

8. Stripe, Voucher and Refund Handling

8.1 Refund calculation

```
current_refundable_cash = sum(cash components linked to the current entitlement)
current_purchased_voucher_restore = sum(purchased-voucher components linked to the current entitlement)
free_discount_cash_refund = 0
```

A full reversal refunds/restores the current components, including documented later top-ups or partial refunds, while preserving the complete original transaction history.

- Allocate order-level discounts and voucher use across seats at purchase time or through a deterministic stored allocation method.
- Never calculate a seat refund by dividing the current order total ad hoc after later changes.
- Store the original pricing snapshot and tax treatment relevant to the transaction.
- Support partial refunds for individual seats within one Stripe payment.

8.2 Semi-automated execution

37. Policy engine calculates outcome and amount.
38. Refund record is created as REQUESTED.
39. Admin sees seat, original payment, voucher components, calculation and reason.
40. Admin confirms execution; system uses a unique idempotency key.
41. Stripe and voucher restoration are processed and logged.
42. Refund remains PROCESSING until provider confirmation/webhook.
43. Brevo confirmation is sent after initiation and may be followed by completion/failure handling.

8.3 Price differences for rebooking

Scenario	Self-service MVP	Admin capability / future
Equivalent configured regular workshop	Direct rebooking	Direct rebooking
Higher-priced non-equivalent target	Not offered in MVP self-service	Collect difference or waive as goodwill
Lower-priced non-equivalent target	Not offered in MVP self-service	Partial refund or documented agreement
Special or partner event	Not offered unless explicitly configured	Any target allowed with explicit price handling

9. Brevo Communication Requirements

- Templates should be generic and dynamic rather than hard-coded to Tempeh/Kombucha/Lacto.
- Every message must include workshop title, date/time, location, number of affected seats, financial outcome and next action.
- Do not promise an exact date on which the bank will display a refund. State that the refund has been initiated and will appear within normal payment-provider processing time.
- Every send must be logged and manually resendable.
- An email failure must not roll back a successful booking/refund, but must surface as an operational exception.

Event	Required customer message
BOOKING_CONFIRMED	Booking details, seats, payment confirmation, cancellation/rebooking summary, terms snapshot/link.
CUSTOMER_REBOOKED	Old and new date, location, seat count, any price handling.
CUSTOMER_CANCELLED_NO_REFUND	Cancellation acknowledged; no refund/rebooking; replacement alternative if still relevant.
REFUND_INITIATED	Affected seat(s), amount, payment method, processing wording.
ORGANISER_CANCELLED	Affected date, reason, secure choice link, rebooking and full refund options.
MAJOR_CHANGE	Change details and choices: accept, rebook, refund.
REPLACEMENT_RECORDED	Updated participant information.
PAYMENT_BOOKING_EXCEPTION	Support message after manual resolution; do not send confusing automatic confirmation.

10. Capacity and Minimum-Participant Logic

Workshop type	Default regular capacity	Default maximum capacity
Lacto-Fermentation	6	10
Tempeh	8	12
Kombucha	8	12

- Values are defaults and must be editable per appointment.
- Customers book automatically only up to the currently released capacity.
- Admins may release additional seats up to maximum capacity after assessing the group and operational conditions.
- Minimum participants default to 4, but this must never trigger automatic cancellation.
- A paid booking above regular capacity but within maximum capacity should be flagged for review rather than automatically refunded.
- Above maximum capacity, the system must not auto-confirm. Admin must either explicitly change maximum capacity or offer rebooking/refund.

`capacity_consuming_entitlements = [ACTIVE, REBOOKING_PENDING_TARGET_HELD]`

`available_online = released_customer_capacity - capacity_consuming_entitlement_count`
`released_customer_capacity <= maximum_capacity`

Exact field names and derived-status implementation are developer choices; capacity changes must be atomic and auditable.

11. Error Handling, Idempotency and Reconciliation

Exception	Required behaviour
Stripe paid, booking missing	Create exception, attempt safe booking recovery, alert admin; do not silently lose payment.
Booking active, payment failed/missing	Block confirmation or mark payment exception; do not count as fully confirmed without defined policy.
Payment/booking double submission	Detect duplicate intent/order; refund erroneous duplicate in full.
Refund webhook delayed/failed	Keep PROCESSING/FAILED, alert admin, permit safe retry with idempotency.
Brevo send failed	Booking action remains valid; flag communication failure and allow resend.
Capacity mismatch	Recalculate from seat states; show discrepancy report before destructive correction.
Voucher mismatch/double use	Lock or transactional redemption; raise exception and preserve transaction history.

Reconciliation requirement: Provide an admin view or scheduled job that compares orders, active seats, Stripe payment/refund states, voucher balances and Brevo send status. Exceptions must be visible without database access.

12. Security, Permissions and Auditability

- Customer management links must be high-entropy, scoped to the specific order/seat and revocable.
- A link may expire technically; the underlying right must not expire solely because the link expired. Support re-issuing links.
- Do not require a customer account for cancellation, rebooking, replacement or organiser-cancellation choice.
- High-risk actions - refund execution, capacity maximum change, voucher balance adjustment - require authenticated admin permissions.
- Audit logs must be append-only from the normal application interface.
- Personal data shown through magic links must be minimised.
- Webhook signatures must be verified, and duplicate events must be safe.

13. Configuration Requirements - Do Not Hard-Code

Configuration	Current value / behaviour
Full-refund threshold	720 hours before original start
Self-rebooking threshold	336 hours before original start

Configuration	Current value / behaviour
Short-notice correction window	24 hours after confirmation, target start at least 48 hours away
Minor/major time-change threshold	60 minutes
Low-participation cancellation latest time	48 hours before start
Minimum participants	4
Self-rebooking count	1
Refund execution mode	Semi-automated, admin confirmation
Default time zone	Europe/Vienna
Current standard rebooking group	Tempeh, Kombucha and Lacto regular workshops
Self-rebooking equivalence/price rule	Configuration-driven; current standard workshops are equivalent when the applicable policy confirms eligibility. No literal price or workshop-name check.

- No literal €99 check in code.
- No workshop-name check for eligibility.
- No global capacity constant shared by all appointment types.
- No assumption that one order equals one seat.
- No assumption that one booking always maps to one date forever; preserve room for course series.

14. MVP Outcome-Based Acceptance Criteria

- ☐ Admins can cancel an appointment, enter reason/internal note, notify all affected customers and see each seat's unresolved choice.
- ☐ A customer can self-rebook one eligible seat once without an account, subject to exact time rules and capacity.
- ☐ A customer can report cancellation inside 14 days and sees an explicit no-refund/no-rebooking warning before confirmation.
- ☐ A replacement participant can be recorded for any seat, including single-seat orders.
- ☐ One seat in a multi-seat order can be cancelled/rebooked/refunded without changing remaining seats.
- ☐ Policy-eligible refunds are calculated per seat and require admin confirmation before Stripe execution.
- ☐ Purchased voucher value is restored correctly; free promotional value is never converted into cash.
- ☐ Admins can move any seat to any appointment/event and explicitly resolve capacity and price differences.
- ☐ Every material action is recorded in an audit trail and triggers the correct Brevo template or a visible communication error.
- ☐ The system stores AGB/cancellation-policy version and acceptance timestamp for every order.
- ☐ Stripe payment, refund, booking and capacity discrepancies are visible in an operational exception view.
- ☐ All thresholds and workshop eligibility groups are configuration-driven.
- ☐ Developers may use a different schema or state model if the proposal preserves all business outcomes, versioning, auditability and reconciliation requirements.
- ☐ Financial calculation after a price-adjusted admin rebooking includes later top-ups or partial refunds and preserves the original ledger/history.
- ☐ Major changes create a per-seat accept/rebook/refund decision flow.
- ☐ Organiser-cancellation rebooking does not consume or reset the customer's one-time voluntary self-rebooking count.
- ☐ A confirmed cancellation is irreversible in customer self-service after the seat is released; replacement must be chosen instead of cancellation.

15. Future Compatibility and Explicitly Deferred Features

Feature	Architecture requirement now	MVP implementation
Different-price customer self-rebooking	Store target price and payment difference; configurable policy	Admin only
Customer credit account	Do not make booking model dependent on accounts	Not required
Goodwill credit	Use auditable voucher/code if needed	Admin only
Waitlist	Seat release events should be emitted	Deferred
Course series / multi-session product	Avoid hard assumption one order = one appointment	Deferred

Feature	Architecture requirement now	MVP implementation
Online workshops	Event type and dynamic communication fields	Supported conceptually
Shipping	Fulfilment method already separated from PICKUP	Keep inactive until operational/legal setup is complete
Automatic refunds	Policy engine and clean refund states	Manual confirmation first

Appendix A - Illustrative Decision Pseudocode

Illustrative only - exact functions and state representation are developer choices.

```
function customerActionOptions(seat, now):
  if seat.appointment.lifecycle == CANCELLED_BY_ORGANISER:
    return [SELECT_ORGANISER_REPLACEMENT, REQUEST_FULL_REFUND]

  if seat.entitlement_status != ACTIVE:
    return permittedActionsForCurrentState(seat)

  hoursToStart = diffHours(seat.appointment.start_at, now, 'Europe/Vienna')

  if seat.self_rebooking_count == 0:
    if hoursToStart >= 720:
      return [REQUEST_FULL_REFUND, SELF_REBOOK, REPLACE_PARTICIPANT]
    if hoursToStart >= 336:
      return [SELF_REBOOK, REPLACE_PARTICIPANT, CANCEL_NO_REFUND]

  if hoursSince(seat.booking_confirmed_at, now) <= 24 and hoursToStart >= 48:
    return [CORRECT_BOOKING_ONCE, REPLACE_PARTICIPANT, CANCEL_NO_REFUND]

  if hoursToStart >= 0:
    return [REPLACE_PARTICIPANT, CANCEL_NO_REFUND]

  return []
```

Note: confirming CANCEL_NO_REFUND releases the entitlement and removes the later self-service replacement option.

Illustrative orchestration only - do not rely on a database transaction spanning Stripe and the local database.

```
function confirmRefundExecution(refundRequest):
  lockAndValidate(refundRequest)
  assert refundRequest.status == REQUESTED
  assert currentUser.hasPermission('refund.execute')

  operation = createOrReuseRefundOperation(
    request=refundRequest,
    idempotencyKey=refundRequest.id,
    status=PROCESSING
  )
  commitLocalState()

  invokeCashRefundIdempotently(operation.cashComponents)
  restorePurchasedVoucherComponentsIdempotently(operation.voucherComponents)

  provider/webhook results update each component to COMPLETED or FAILED
  aggregate refund status is derived from component states
  discrepancies remain visible for reconciliation and safe retry
```

Illustrative only.

```
function canSelfRebook(sourceSeat, targetAppointment, now):
  return (
    sourceSeat.entitlement_status == ACTIVE
    and sourceSeat.self_rebooking_count == 0
    and sourceSeat.appointment.start_at - now >= configured.self_rebook_threshold
    and targetAppointment.isCustomerBookable()
    and configuredRebookingPolicy.allows(sourceSeat, targetAppointment)
    and targetAppointment.hasCustomerBookableCapacity()
  )
```

The policy may use group, price category, effective price or another configuration; it must not rely on literal workshop names or €99.

Appendix B - Event and Email Trigger Matrix

Domain event	Side effects	Must be idempotent?
PAYMENT_SUCCEEDED	Create/confirm order-booking if valid; update payment; trigger booking email	Yes
BOOKING_CONFIRMED	Consume capacity; write terms snapshot; queue Brevo confirmation	Yes
CUSTOMER_REBOOK_COMPLETED	Release source seat; consume target; link records; email	Yes
CUSTOMER_CANCELLED_NO_REFUND	End entitlement and release capacity; record no financial action; email. This is final in self-service.	Yes
REFUND_REQUESTED	Create refund record; admin task/notification	Yes
REFUND_EXECUTION_CONFIRMED	Stripe refund + voucher restoration + audit	Yes
REFUND_UPDATED_BY_STRIPE	Update provider status; send/follow up on failure	Yes
APPOINTMENT_CANCELLED_BY_ORGANISER	Mark affected entitlements pending choice; send choice links; do not consume voluntary self-rebooking allowance.	Yes
PARTICIPANT_REPLACED	Update participant details; email if configured	Yes
BREVO_SEND_FAILED	Create visible operational exception; allow resend	Yes
APPOINTMENT_MAJOR_CHANGE	Mark affected entitlements pending accept/rebook/refund choice; notify customers.	Yes

Implementation handover: Before coding, Rafaela and Ala'a should map the required business outcomes to the existing Payload collections and Stripe/Brevo flows, identify migration needs for current bookings, and return a proposed schema plus state-transition diagram. The field names, statuses and pseudocode in this document are references, not mandatory structures. Where the current codebase supports a simpler or safer solution, preserve the AGB-aligned outcomes, customer rights, auditability and reconciliation, and document the proposed deviation for Fermentfreude approval.